Compositional Concept Synthesis via CLIP-Guided Primitive Stitching

April 11, 2025

Paolo Cursi Pietro Signorino

Abstract

In this paper we explore the task of creating a composite image described by a natural language prompt from a set of given primitives, such as combining "person" and "umbrella" to form "a person under an umbrella." Each primitive is represented as a visual element and a composition function arranges them spatially to form a complete scene. We use CLIP to guide this composition by evaluating how well the rasterized image of the composed scene aligns with the target natural language description. The optimization objective is to maximize the cosine similarity between the image embedding and the text embedding produced by CLIP.

1. Introduction

Our work focuses on the synthesis of complex composite image by stitching together a set of predefined primitives. Each target concept is associated with a collection of k primitives, which are arranged to maximize the cosine similarity between the generated image embedding and the corresponding text embedding, both produced by CLIP (Radford et al., 2021).

We tried to solve this challenging task by using three different optimization methods which did not work as expected. This happened because of a challenging loss function landscape, which exhibited many local maxima and impeded the optimization process from converging to the global optimum. We performed an investigation of this problem, which lead us to discover some interesting CLIP limitations that we will talk about in the next sections.

2. Related Work

We based our work on the paper *Modern Evolution Strategies for Creativity: Fitting Concrete Images and Abstract Concepts* (Tian & Ha, 2022) (ES-CLIP).

ES-CLIP presents a method that leverages evolutionary strategies to optimize image generation based on CLIP's joint embedding space. An element in the *population* is composed of a set of overlapping triangles each one with its own vertex position and color. These features are optimized to maximize the cosine similarity between the CLIP embedding of the triangles rendered on a canvas and the CLIP embedding of a target prompt.

Inspired by this work, we want to use CLIP's embedding space to compose visual scenes from a set of semantic primitives, instead of triangles.

3. Method

In order to stitch together k primitives to form a scene, we need to define a few components: the parameter to be optimized and the optimization strategy.

3.1. Parameters & Pipeline

We represent the parameters as a matrix of shape $(k, 5)$ and each entry is bounded to be in the $[0, 1]$ range. Each row corresponds to one primitive and the five parameters describe its spatial attributes.

- **x, y**: the position of the top-right corner of the primitive with respect to the canvas size;
- **scale**: the relative size of the primitive (i.e. a percentage of the full primitive size);
- **rotation**: the angle of rotation $r \in [0, 365]$ degrees normalized;
- **layer**: the rendering order (i.e., which primitives appear in front). We order the rasterization process, such that primitives with smaller layer parameter will be rasterized first, hence be in the back.

We created a `rasterize` function, which uses these parameters and the primitives to compose the scene and rasterizes it into an image. This image is then fed into CLIP's

Email: Paolo Cursi <cursi.2155622@studenti.uniroma1.it>, Pietro Signorino <signorino.2149741@studenti.uniroma1.it>.

Deep Learning and Applied AI 2024, Sapienza University of Rome, 2nd semester a.y. 2023/2024.

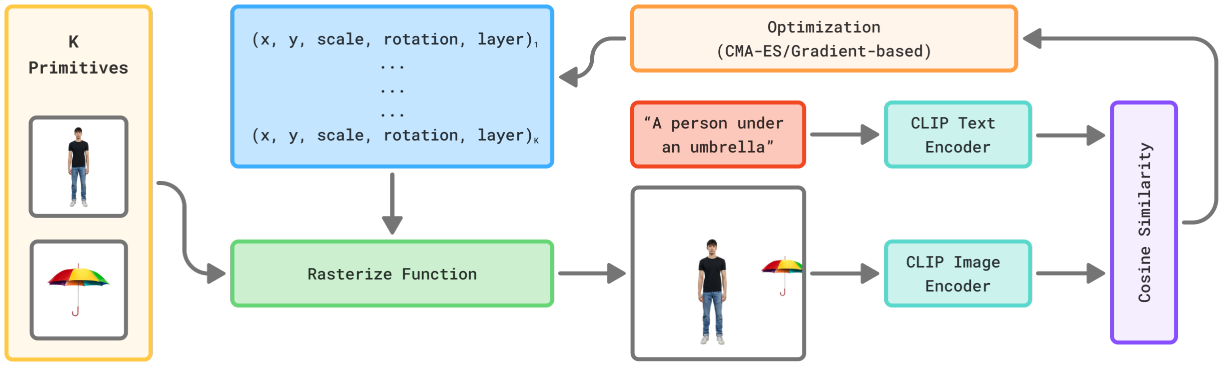


Figure 1. The whole pipeline of our method. We want the k (in this case 2) primitives shown in the *yellow* box, to be rasterized to have a similar CLIP embedding as the prompt, shown in the *red* box.

image encoder to compute its embedding. The cosine similarity between this image embedding and the text embedding of the target prompt serves as the objective function. An overview of this pipeline is shown in Figure 1.

3.2. PGPE Optimization

The first alghoritm we tried is PGPE (Probabilistic Gradient Population Evolution) (Toklu et al., 2020), the one used in the ES-CLIP paper. This algorithm is a gradient-free optimization method that uses a population of candidate solutions to explore the search space. It maximizes the target function so we used a loss $\mathcal{L} = \cos\_\text{sim}(\text{img}, \text{prompt})$, i.e. the cosine similarity between the image embedding and the text embedding of the target prompt. This method did not yield good results as in the implementation of the ES-CLIP paper. We belive that the reason are *sparse rewards*, we will discuss this further in the next sections.

3.3. CMA-ES Optimization

To overcome the limitations of PGPE, we decided to use a more advanced Evolution Strategy implementation, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) (Hansen & Ostermeier, 1996). Compared to PGPE, CMA-ES uses a more sophisticated sampling algorithm, which prioritize directions where improvement has been observed updating a multivariate Gaussian distribution. It also minimizes the target function so we used a loss $\mathcal{L} = 1 - \cos\_\text{sim}(\text{img}, \text{prompt})$. This methods typically works better with sparse rewards, but in our case it did not produce significant improvements, just slightly better results compared to PGPE.

3.4. Gradient-Based Approach

In parallel, we also experimented with a gradient-based optimization method. We created a new rasterization function which is differentiable and has some limitations: rotating the primitives was pretty hard, so we decided to implment it only if a preliminary test would have been promising. This function allowed us to compute gradients of the cosine similarity loss with respect to the primitive parameters and apply gradient descent to improve the layout. Unfortunately it do not yielded good results as well.

4. Approach Limitations

In this section we will discuss the problems we encountered during the optimization process and try to explain why they cannot be easly overcomed.

4.1. The Sparse Rewards Problem

As stated earlier, we believe one issue with our approach is the well-known sparse rewards problem. This common challenge in Reinforcement Learning arises because even slight variations around a given point in the solution space can produce inconsistent reward values, making it difficult to optimize and causing the gradient to sometimes lead to suboptimal solutions. If the best solution is far from the sampled points, the algorithm will not be able to reach it, since intermediate solutions do not provide an incremental improvement in the reward. Therefore the loss landscape is very noisy, often inconsistent, has many local optima and it is has a lot of ”flat regions” (i.e., regions where the loss does not change much).

4.2. The CLIP problem

Another important problem we faced has to do with the CLIP embedding space. Given the embedding of a prompt, it can happen that it has an higher cosine similarity with a meaningless composition of primitives, instead of a manually crafted one. An example of this can be seen in Figure 3: the prompt used for the similarity computation is *A car on the road*. The ground truth image (a) has a cosine

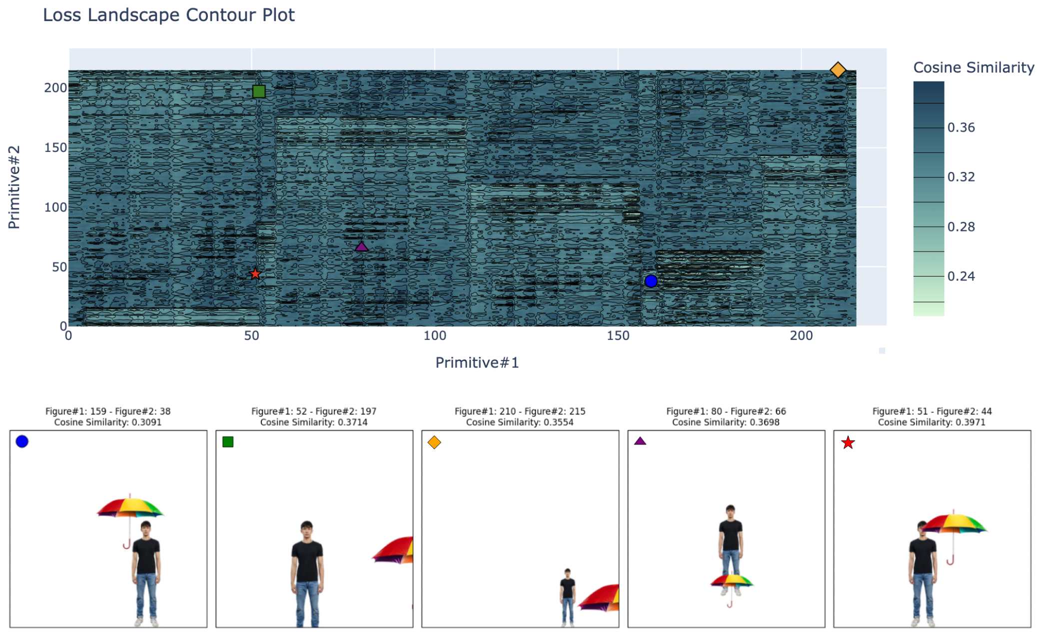


Figure 2. Loss landscape of the function for the prompt "A person under an umbrella". The red star indicates the maxima.

similarity score of 0.2675 while the random one (b) has a score of 0.2755. This is a big problem, because we cannot rely on the cosine similarity to be an effective measure of how good our solution is.

(a) (b)

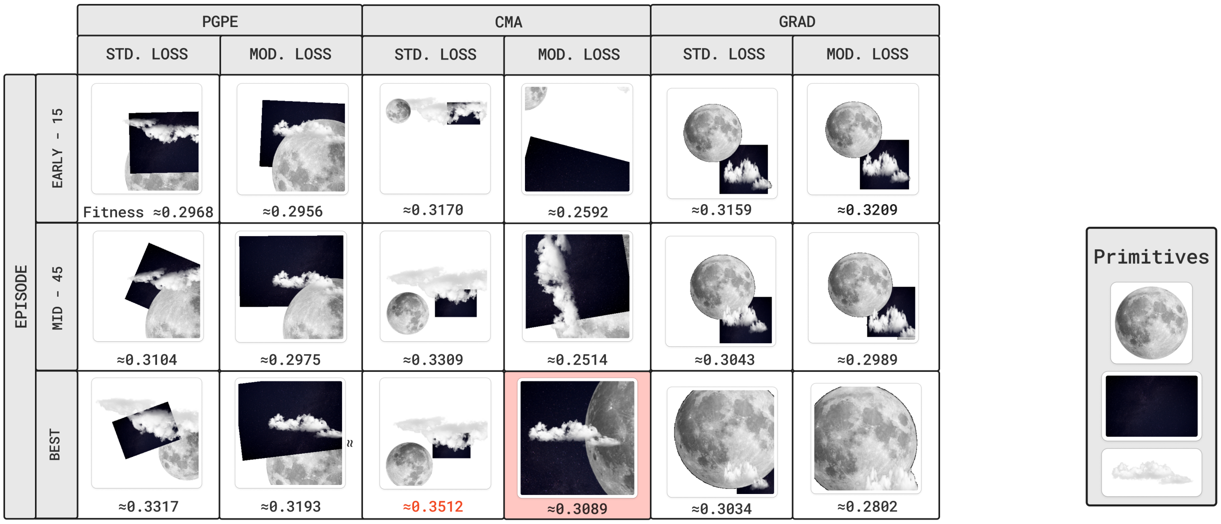
Figure 3. Examples demonstrating CLIP's inconsistency.
(a) The ground truth image
(b) The generated one

4.3. More on the Loss Landscape

After the previous observations, we decided to explore the loss function's landscape. We choose a simple example *A person under un umbrella*, that has two primitives: *person* and *umbrella*. We embedded every combination of $(x, y, scale)$ parameters with values $0.3, 0.38, 0.46, 0.54, 0.62, 0.7$ for both primitives, while keeping the rotation and layer fixed. In Figure 2 we plotted the contour plot of the loss landscape: even if the image with maximum prompt-similarity is well composed, meaningless compositions (e.g. purple triangle) have higher prompt-similarity than a meaningful one (blue circle). Furthermore, we calculated the *Coefficient of Variation* (CV) of the all the loss values computed for this experiment. This coefficient is a measure of the relative variability of a distribution, a low value indicates that the values are close to the mean. We found a CV of 0.0583 (or $\approx 5\%$), confirming that the loss function is pretty flat.

5. Results & Improvement

In order to deal with the problems described, we tried to change the loss function. In a particular use-case, where a background image is present we can add a term to the loss function, leading us to: $\mathcal{L}' = \text{cos\_sim(img, prompt)} * (1 - \gamma) + \text{T} * \gamma$. The new term T is the count of empty pixels in the canvas, normalized to be in $[0, 1]$. This term, weighted for $\gamma = 0.2$ will change the loss landscape, because rasterization with "empty" canvases will be penalized. By adding this term, only for rasterization that require a background we obtained some good results when using CMA-ES and this is not trivial at all. An example is shown in Figure 4, where we plot rasterizations done with all three methods, using the standard (L) and this new $(\mathcal{L}')$ loss. The parameters where also changed, in particular **x, y** where bounded to be in the $[-0.5, 1]$ range and the **scale** to be in $[0.2, 1.5]$, in order for the optimization method to be able to make the background image bigger and cover the whole screen. Other examples are shown in the project's folder.



(a) (b)

Figure 4. Results for each method with both the original and the modified loss function, with prompt "A cloudy night sky with a full moon". The image with the red background is visually the best result, but it has less prompt-similarity (fitness) than the image with the fitness written in red.

6. Conclusions

Despite implementing several optimization strategies, our results fell short of expectations. Our investigation revealed significant challenges that impeded the optimization process: sparse rewards in the solution space making convergence difficult, inconsistencies in CLIP's ability to discriminate between semantically meaningful and random arrangements and a highly non-convex loss landscape with numerous local optima and flat regions.

These findings suggest that future work should focus on developing more robust optimization algorithms specifically designed to navigate sparse reward landscapes. Additionally, incorporating auxiliary reward signals or hierarchical optimization approaches might help guide the search process more effectively. Perhaps changing the embedding model, or fine-tuning CLIP to be "semantically reliable" could also significantly improve the performance of this approach.

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